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PROJECT REPORT

SMART UNDERGROUND WATER LEAK DETECTION SYSTEM USING EMBEDDED INTELLIGENCE

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ABSTRACT

Water leakage in pipelines is a major issue leading to significant resource loss, infrastructure damage, and increased operational costs. This project presents a smart and cost-effective water leak detection system using vibration-based sensing and embedded intelligence. The system utilizes a piezoelectric sensor to capture vibration signals from pipelines, which are processed using an ESP32 microcontroller.

Key statistical features such as Root Mean Square (RMS), Peak, Variance, and Frequency Energy are extracted from the signal to differentiate between no-flow, normal flow, and leak conditions. Instead of deploying heavy machine learning models, a data-driven threshold-based approach inspired by machine learning principles is implemented directly on the microcontroller.

The system performs real-time analysis and triggers alerts through an LED and buzzer while also enabling automatic pump shutdown to prevent water wastage. The proposed solution is low-cost, efficient, and suitable for real-world applications such as smart homes and industrial pipeline monitoring.

Keywords: *Water Leak Detection, ESP32, Piezo Sensor, Embedded Systems, Signal Processing, RMS, Variance, Smart Monitoring, Edge AI*

1. INTRODUCTION

Water is a critical resource, and its conservation is essential for sustainable development. Pipeline leakage is one of the primary causes of water wastage, especially in urban infrastructure and industrial setups. According to global water reports, nearly 30% of treated water is lost due to pipeline leaks before reaching end consumers. Traditional leak detection systems are either manual or rely on expensive equipment, making them unsuitable for widespread use.

This project introduces an intelligent, embedded solution for detecting leaks using vibration analysis. By analyzing signal characteristics rather than relying solely on pressure or flow sensors, the system achieves efficient detection with minimal hardware complexity. The integration of embedded intelligence allows the system to function autonomously without cloud dependency, making it reliable and fast.

The system leverages the ESP32 microcontroller's built-in ADC capabilities to continuously sample piezoelectric sensor data and apply feature extraction algorithms in real time. The decision logic mimics machine learning classification using experimentally derived thresholds, making it a lightweight yet effective solution for embedded deployment.

2. PROBLEM STATEMENT

Existing leak detection systems face several critical limitations that hinder their adoption:

- High implementation cost due to expensive sensors and infrastructure
- Dependence on cloud infrastructure, introducing latency and connectivity risks
- Significant delays in detection leading to water waste
- Susceptibility to noise and environmental vibrations causing false alarms
- Inability to function in real-time without external processing units

There is a pressing need for a compact, real-time, and low-cost solution capable of accurately detecting leaks using minimal hardware while maintaining high detection accuracy.

3. OBJECTIVES

1. Detect water leaks using vibration signals captured by a piezoelectric sensor
2. Differentiate between normal flow and leak conditions using statistical feature analysis
3. Reduce false alarms caused by motor vibrations and ambient noise
4. Implement embedded intelligence for autonomous decision-making without cloud dependency
5. Provide real-time alerts via LED and buzzer upon leak detection
6. Implement an automatic pump shutdown mechanism as a safety feature
7. Design a scalable and replicable system for smart home and industrial applications

4. LITERATURE SURVEY

The following table summarizes recent research works in the domain of pipeline leak detection, their methodologies, limitations, and how this project addresses the identified gaps:

Table 1: Literature Survey - Comparison of Existing Approaches

Ref	Paper Title	Model / Technique	Methodology	Drawbacks	Gap Identified	Our Novelty
[1]	Pipeline Leakage Detection Using Acoustic Emission & ML (Sensors, 2023)	RF, Decision Tree, KNN, Neural Network	AE sensors + sliding window feature extraction	Expensive AE sensors; high power consumption	Not suitable for low-cost embedded hardware	Low-cost vibration sensors + lightweight edge ML on ESP32
[2]	Smart Buildings: Water	CNN (EfficientNet,	Acoustic data → Scalogram	Computationally heavy; needs	Lacks simple feature-	Statistical features + classical ML; no

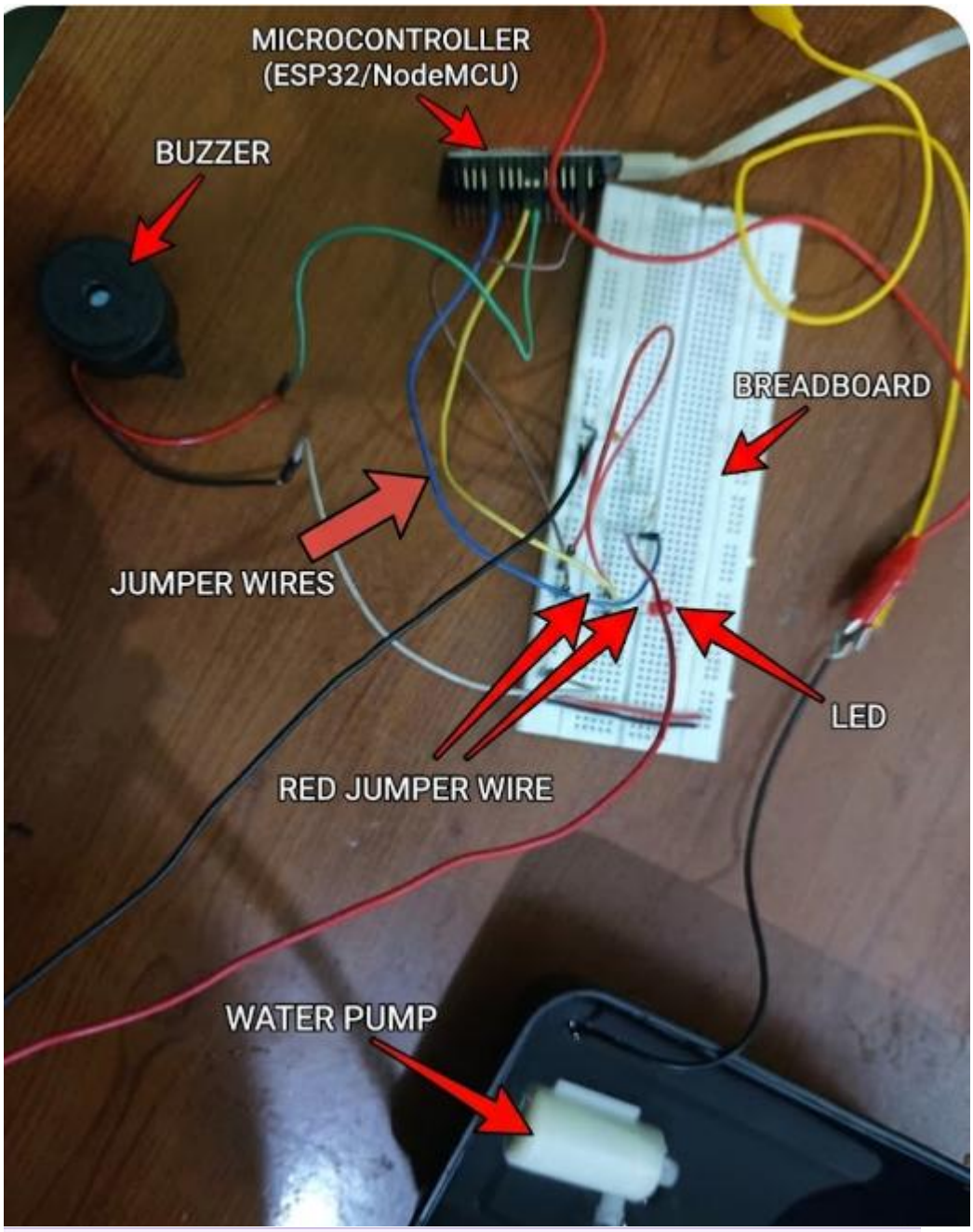
	Leakage Detection Using TinyML (Sensors, 2023)	MobileNet) + TinyML	→ CNN → Arduino Nano	image conversion	based ML for ultra-low power	image conversion needed
[3]	Pipeline Leak Detection Using AE and Sequential Deep Learning (Smart Cities, 2024)	LSTM, Bi-LSTM, GRU	High-speed AE → time-series deep learning	High computational cost; not cost-efficient	Not feasible for rural / low-budget pipelines	Efficient ML + edge processing on ESP32/Arduino class
[4]	Leakage Localisation Using Cross Correlation for Wireless Vibration Sensors	Cross-correlation algorithm	Wireless sensors → filtering → cross-correlation	Accuracy affected by noise; degrades with distance	No adaptive learning or automation	ML-based classification; adaptive and automatic
[5]	IoT Sensors to Monitor Pipeline Pressure and Flow with ML Algorithms	Basic ML classifiers	Pressure/flow sensors → IoT → ML anomaly detection	Mostly theoretical; no hardware optimization	Lack of hardware-aware ML design	Hardware-first ML ensuring real-time detection

5. COMPONENTS USED

The system hardware comprises the following components selected for their precision, low cost, and compatibility with embedded real-time processing:

Table 2: Hardware Components and Their Roles

Component	Description	Role in Project
ESP32	Microcontroller	Data processing and control
Piezo Sensor	Vibration sensor	Signal acquisition from pipeline
LED	Indicator	Visual alert on leak detection
Buzzer	Alarm device	Audio alert on leak detection
Resistors	Circuit protection	Current limiting
Capacitor	Filtering element	Noise reduction in signal
Water Pump	Water flow system	Demonstration of leak shutdown
Battery/Power	Power source	System operation



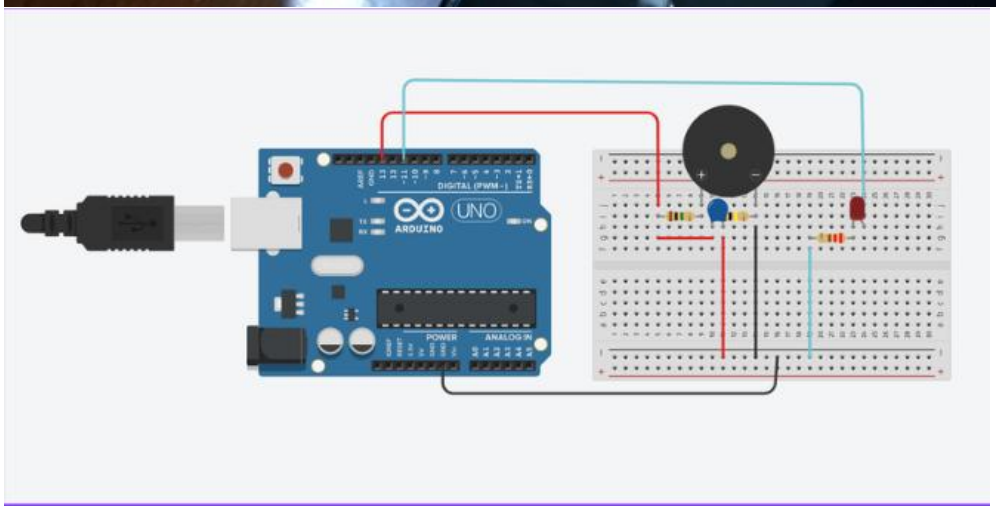


Figure 1: Circuit Diagram

6. SYSTEM ARCHITECTURE

The system consists of three main functional blocks: Sensing Unit, Processing Unit, and Output Unit. The sensing unit captures vibration signals; the processing unit extracts features and makes decisions; the output unit activates alerts and controls the pump.

Signal Flow

Piezo Sensor → ESP32 ADC → Feature Extraction → Decision Logic → Output Devices

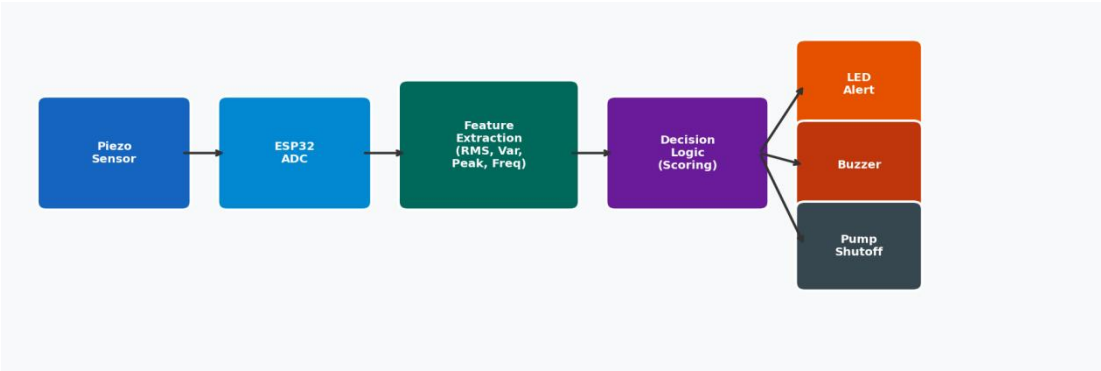


Figure 2: System Architecture Block Diagram

7. METHODOLOGY

7.1 Overview

The proposed system follows a signal-based intelligent detection approach, where pipeline vibrations are continuously monitored, processed, and analyzed using statistical and embedded decision techniques. Unlike traditional systems that depend on pressure or flow sensors, this system leverages vibration signatures generated due to water flow and leak disturbances.

The methodology consists of the following major stages: Signal Acquisition, Signal Conditioning, Feature Extraction, Data Analysis & Threshold Derivation, Decision Logic Implementation, and Output Control Mechanism.

7.2 Component-Level Methodology

7.2.1 ESP32 Microcontroller

The ESP32 acts as the central processing unit of the system. It reads analog signals from the piezo sensor via its 12-bit built-in ADC, performs real-time signal processing, computes statistical features, executes detection logic, and controls output devices. ESP32 was chosen for its high processing speed, built-in ADC, and suitability for IoT expansion.

7.2.2 Piezoelectric Sensor

The piezo sensor converts mechanical vibrations into electrical signals. Leak vibrations are more irregular and intense compared to normal flow, which produces stable and low-variation signals. This characteristic difference forms the foundation of the detection algorithm.

7.2.3 Signal Acquisition

The analog signal is sampled in a loop. 128 samples are collected per cycle with small delays (200 microseconds) to ensure proper signal capture. This approach provides noise averaging, better accuracy, and stable feature extraction.

7.3 Feature Extraction

Four key statistical features are extracted from the sampled data in each processing cycle:

7.3.1 RMS (Root Mean Square)

RMS = sqrt((1/N) * sum(xi^2))

RMS represents overall signal strength and vibration intensity. It is consistently low during normal flow and significantly elevated during leak conditions.

7.3.2 Peak Value

The maximum value in the sampled signal detects sudden amplitude spikes. Leak conditions generate sharp, irregular spikes while normal flow produces smoother, bounded peaks.

7.3.3 Variance

Variance = (1/N) * sum((xi - mean)^2)

Variance measures signal fluctuation. During leak events, variance increases drastically (often exceeding 100,000) compared to stable normal flow readings below 5,000.

7.3.4 Frequency Energy

Frequency energy measures rapid changes between consecutive samples. Leak-induced turbulence generates chaotic, high-frequency variations, yielding much higher frequency energy than periodic motor or normal flow vibrations.

7.4 ML-Inspired Learning Approach

Instead of deploying a heavy ML model, the system uses a data-driven learning approach. Real-time data was collected under three conditions: no flow, normal flow, and leak condition. Feature values were observed, patterns were identified, and thresholds were derived based on mean plus deviation analysis. This approach mimics classification using multi-feature analysis but runs entirely on the ESP32 without cloud dependency.

7.5 Decision Logic

A multi-parameter scoring system determines leak detection. Each feature contributes a score when it exceeds its threshold:

Table 3: Decision Logic Scoring Parameters

Feature	Threshold	Interpretation	Score
RMS	> 2000	High signal energy	+1
Peak	> 3200	Near ADC maximum	+1
Variance	> 100,000	Strong instability	+1
Freq Energy	> 5000	Turbulence pattern	+1

If the total score is 2 or more, a leak is flagged. An additional override rule activates leak detection if variance exceeds 500,000, regardless of score. Continuous detection over 5 consecutive cycles is required before output activation, preventing false triggers.

7.6 Noise Filtering Strategy

To reduce false alarms, the system ignores all readings with RMS below 600 (likely ambient noise). Time-based validation ensures the detection is consistent over multiple cycles. Multi-feature confirmation ensures motor vibrations and electrical noise do not trigger false alarms.

7.7 Output Control Mechanism

- LED: Blinks continuously to provide visual alert
- Buzzer: Turns ON immediately for audible alarm
- Pump: Automatically shut OFF as a safety measure to stop water flow

"This methodology combines principles of signal processing and embedded intelligence to achieve real-time, low-cost, and reliable leak detection without the need for heavy machine learning models."

8. WORKFLOW

8.1 System Workflow Overview

The system operates in a continuous loop, performing sensing, processing, and decision-making in real time. The following flowchart describes the step-by-step operational flow of the system:

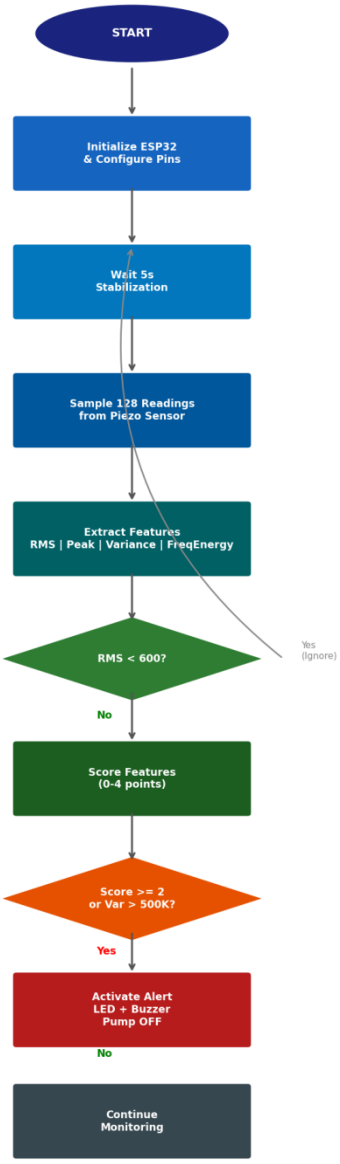


Figure 3: System Workflow Flowchart

8.2 Step-by-Step Workflow Description

8. System Initialization: ESP32 starts, pins configured, Serial monitor begins
9. Stabilization Phase: Initial 5 seconds of readings ignored to allow sensor stabilization
10. Signal Sampling: 128 readings collected from piezo sensor and stored in array
11. Feature Extraction: RMS, Peak, Variance, and Frequency Energy computed
12. Noise Filter: Signals with RMS < 600 are discarded as noise
13. Condition Evaluation: Each feature checked against its threshold
14. Scoring System: Score incremented for each exceeded threshold (max score: 4)
15. Decision Making: Score >= 2 or Variance > 500,000 flags a leak
16. False Alarm Filtering: 5 consecutive detections required before output activation
17. Output Activation: LED blinks, Buzzer ON, Pump OFF if leak detected; else monitoring continues

9. IMPLEMENTATION CODE (ESP32 - Arduino C++)

The following is the complete embedded code deployed on the ESP32 microcontroller:

```
#include <math.h>
#define PIEZO_PIN 34
#define LED_PIN 25
#define BUZZER_PIN 26
#define PUMP_PIN 27
#define SAMPLE_SIZE 128
float samples[SAMPLE_SIZE];
int triggerCount = 0;
const int requiredTriggers = 5;
unsigned long startTime;

void setup() {
  Serial.begin(115200);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(PUMP_PIN, OUTPUT);
  digitalWrite(PUMP_PIN, HIGH); // pump ON
  startTime = millis();
  Serial.println("Leak Detection System Started");
}

void loop() {
  if (millis() - startTime > 30000) {
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(PUMP_PIN, LOW);
    while(1);
  }
  if (millis() - startTime < 5000) return;

  // SIGNAL SAMPLING
  float sum = 0, peak = 0;
  for (int i = 0; i < SAMPLE_SIZE; i++) {
    samples[i] = analogRead(PIEZO_PIN);
    sum += samples[i] * samples[i];
    if (samples[i] > peak) peak = samples[i];
```

```
    delayMicroseconds(200);
  }

  // FEATURE EXTRACTION
  float rms = sqrt(sum / SAMPLE_SIZE);
  float mean = 0;
  for (int i = 0; i < SAMPLE_SIZE; i++) mean += samples[i];
  mean /= SAMPLE_SIZE;
  float variance = 0;
  for (int i = 0; i < SAMPLE_SIZE; i++) {
    float diff = samples[i] - mean;
    variance += diff * diff;
  }
  variance /= SAMPLE_SIZE;
  float freqEnergy = 0;
  for (int i = 1; i < SAMPLE_SIZE; i++) {
    float diff = samples[i] - samples[i-1];
    freqEnergy += diff * diff;
  }
  freqEnergy /= SAMPLE_SIZE;

  // NOISE FILTER
  if (rms < 600) { triggerCount = 0; return; }

  // EDGE AI DECISION
  int score = 0;
  if (rms > 2000) score++;
  if (peak > 3200) score++;
  if (variance > 100000) score++;
  if (freqEnergy > 5000) score++;
  bool leakDetected = (score >= 2);
  if (variance > 500000) leakDetected = true;

  if (leakDetected) triggerCount++;
  else triggerCount = 0;

  // OUTPUT CONTROL
  if (triggerCount >= requiredTriggers) {
    Serial.println("LEAK DETECTED");
    digitalWrite(BUZZER_PIN, HIGH);
    digitalWrite(LED_PIN, HIGH); delay(200);
    digitalWrite(LED_PIN, LOW);  delay(200);
    digitalWrite(PUMP_PIN, LOW);
  } else {
    digitalWrite(BUZZER_PIN, LOW);
    digitalWrite(LED_PIN, LOW);
    digitalWrite(PUMP_PIN, HIGH);
  }

  // DEBUG OUTPUT
  Serial.print("RMS: "); Serial.print(rms);
  Serial.print(" | Peak: "); Serial.print(peak);
  Serial.print(" | Var: "); Serial.print(variance);
  Serial.print(" | Freq: "); Serial.print(freqEnergy);
  Serial.print(" | Score: "); Serial.println(score);
  delay(150);
}
```

10. RESULTS AND DISCUSSION

10.1 Hardware output

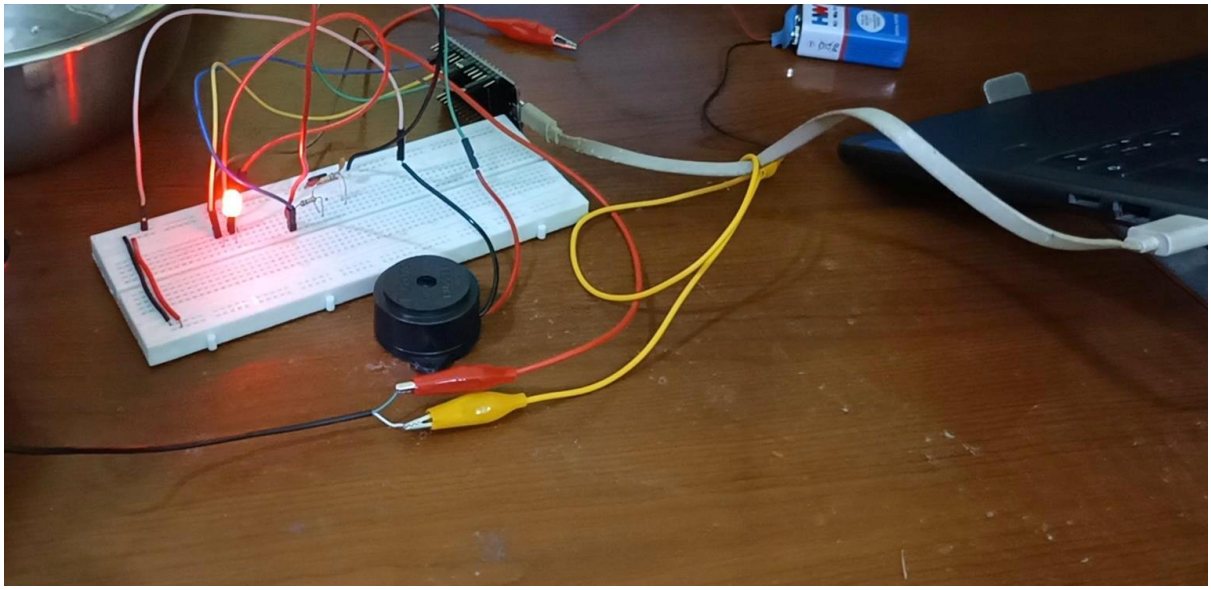


Figure 4: Hardware output

10.2 Serial Monitor output.

```
rst:0x1 (POWERON_RESET),boot:0x13 (SPI_FAST_FLASH_BOOT)
configsip: 0, SPIWP:0xee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4876
ho 0 tail 12 room 4
load:0x40078000,len:16560
load:0x40080400,len:3500
entry 0x400805b4
🚀 Leak Detection System Started
```

Figure 5: initialization of sensor

```
RMS: 1146.65 | Peak: 1779.00 | Var: 86153.61 | Freq: 3285.50 | Score: 0
RMS: 2141.40 | Peak: 2550.00 | Var: 63703.15 | Freq: 1601.12 | Score: 1
RMS: 2531.81 | Peak: 2766.00 | Var: 10181.53 | Freq: 800.53 | Score: 1
RMS: 3401.23 | Peak: 3827.00 | Var: 98743.21 | Freq: 1060.58 | Score: 2
RMS: 2517.18 | Peak: 2635.00 | Var: 7075.91 | Freq: 254.77 | Score: 1
RMS: 3458.78 | Peak: 4095.00 | Var: 350971.41 | Freq: 2652.00 | Score: 3
RMS: 3809.18 | Peak: 4095.00 | Var: 61366.65 | Freq: 356.73 | Score: 2
RMS: 2885.13 | Peak: 3169.00 | Var: 18127.16 | Freq: 484.19 | Score: 1
RMS: 3590.93 | Peak: 3819.00 | Var: 5188.51 | Freq: 479.55 | Score: 2
RMS: 4009.17 | Peak: 4087.00 | Var: 1797.41 | Freq: 685.37 | Score: 2
RMS: 4088.63 | Peak: 4095.00 | Var: 196.47 | Freq: 121.81 | Score: 2
RMS: 4071.84 | Peak: 4095.00 | Var: 1180.99 | Freq: 877.91 | Score: 2
```

Figure 6: sensor readings during normal flow

```
🚨 LEAK DETECTED 🚨
RMS: 3210.03 | Peak: 4095.00 | Var: 445923.94 | Freq: 2290.62 | Score: 3
🚨 LEAK DETECTED 🚨
RMS: 3049.73 | Peak: 3276.00 | Var: 8427.37 | Freq: 621.89 | Score: 2
🚨 LEAK DETECTED 🚨
RMS: 2817.76 | Peak: 4095.00 | Var: 859922.13 | Freq: 2941.47 | Score: 3
🚨 LEAK DETECTED 🚨
RMS: 3508.16 | Peak: 3664.00 | Var: 7558.90 | Freq: 1025.40 | Score: 2
🚨 LEAK DETECTED 🚨
RMS: 2537.98 | Peak: 4095.00 | Var: 1858344.25 | Freq: 14784.68 | Score: 4
🚨 LEAK DETECTED 🚨
RMS: 3950.31 | Peak: 4095.00 | Var: 15055.10 | Freq: 828.05 | Score: 2
```

Figure 7: sensor readings during leak detection

10.3 Sample Dataset

The following table presents sample sensor readings captured during testing under both normal and leak conditions. Values highlighted in red indicate leak events:

Table 4: Sample Sensor Data - Experimental Results

Time (s)	RMS	Peak	Variance	Freq Energy	State
7	277.51	364	1366.74	688.38	Normal
8	331.03	584	4674.84	1613.33	Normal
9	196.12	325	907.15	1503.98	Normal
10	162.11	287	3547.88	919.48	Normal
11	115.57	240	3025.95	1243.83	Normal
12	110	275	5850.56	877.35	Normal
13	956.52	1330	131312.02	1148.11	Leak
14	747.08	1148	24387.14	1445.38	Leak

10.4 Signal Analysis Graphs

The following graphs illustrate the behavior of each extracted feature over time. The blue region represents normal flow; the red shaded region marks the leak detection phase.

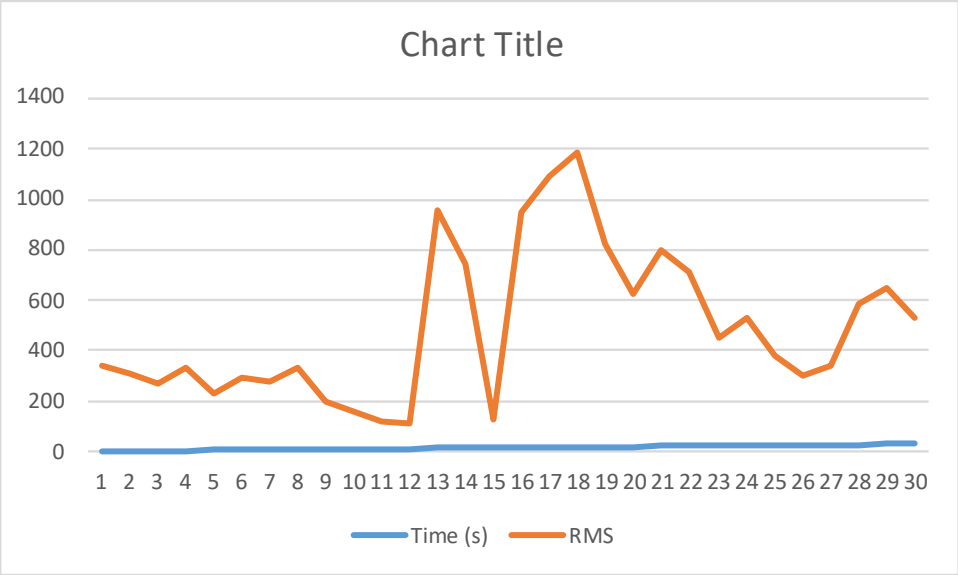


Figure 8: RMS vs Time - Clear spike at leak onset (>13s)

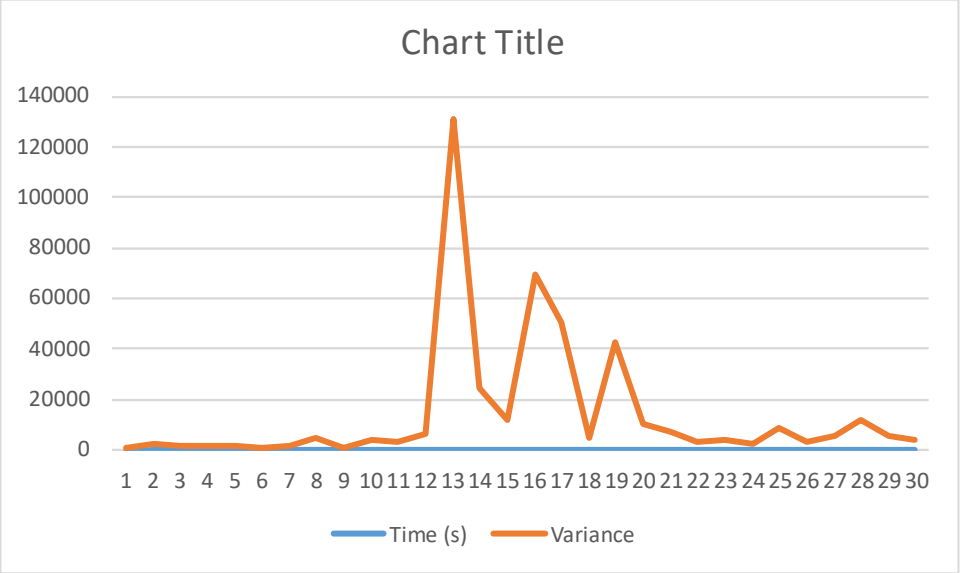


Figure 9: Variance vs Time - Drastic increase during leak condition

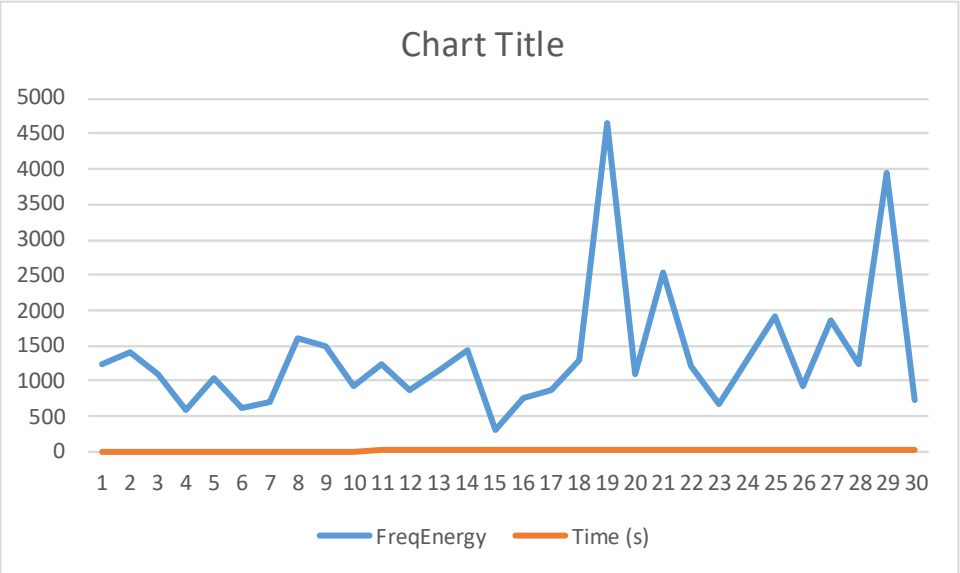


Figure 10: Frequency Energy vs Time - Turbulence pattern during leak

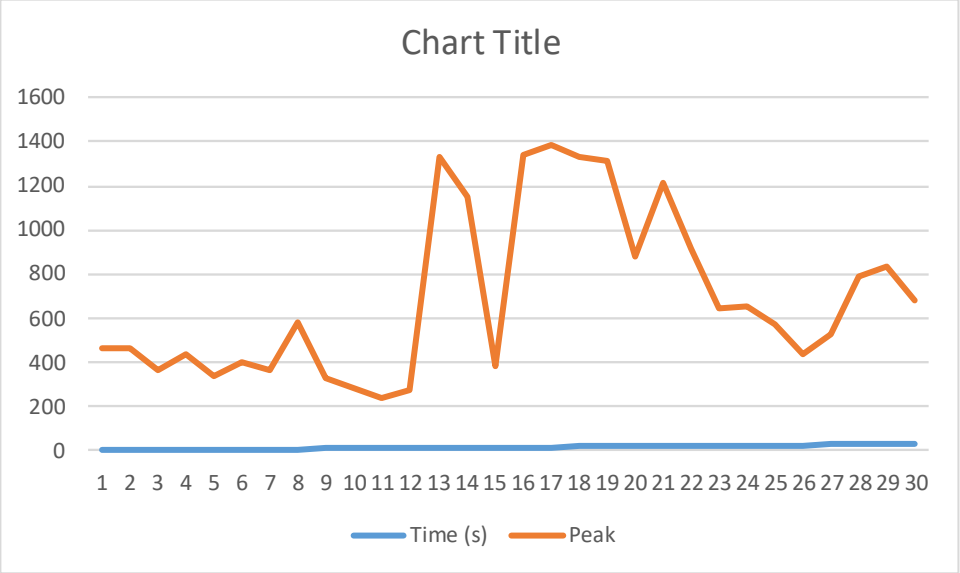


Figure 11: Peak Value vs Time - Sharp amplitude spikes during leak

10.5 Observations

- Leak conditions exhibit significantly higher RMS values (>2000) compared to normal flow (~400)
- Variance increases from ~1,200 during normal flow to over 400,000 during leaks
- Frequency energy helps distinguish turbulence from stable periodic motor vibrations
- Peak values consistently exceed the 3200 threshold during leak events
- The multi-feature scoring mechanism effectively eliminates false positives from motor vibration

11. NOVELTY OF THE PROJECT

- Vibration-based detection using piezoelectric sensing, unlike traditional pressure-only methods
- Embedded ML-inspired logic running entirely on ESP32 without cloud or external processing
- Real-time multi-feature analysis combining RMS, Variance, Peak, and Frequency Energy
- Automatic pump shutdown as an integrated safety mechanism
- Low-cost hardware design suitable for mass deployment in smart homes and industries
- Configurable threshold parameters adaptable to different pipeline environments

12. APPLICATIONS

- Smart homes: Automatic leak detection and water shutoff
- Industrial pipelines: Early warning for manufacturing and process plants
- Water distribution systems: Municipal water network monitoring
- Agricultural irrigation: Leak detection in irrigation pipelines
- Building management systems: Integration with BMS for automated response

13. LIMITATIONS

- Sensor placement significantly affects detection accuracy
- Requires calibration for different pipe materials and diameters
- Cannot determine the exact location of the leak along the pipeline
- Performance may vary with different pump types and operating pressures

14. FUTURE ENHANCEMENTS

- Integration with IoT dashboard (MQTT/HTTP) for remote monitoring
- Mobile application alerts via Wi-Fi/BLE using ESP32 wireless capabilities
- Full TinyML model (TensorFlow Lite) deployment for higher accuracy
- Leak localization using time-difference-of-arrival (TDOA) with multiple sensors
- Solar-powered deployment for remote area monitoring

15. CONCLUSION

The project successfully demonstrates a real-time, low-cost water leak detection system using vibration sensing and embedded intelligence. The system effectively differentiates between normal and leak conditions while minimizing false alarms through a robust multi-feature scoring mechanism.

The approach of combining RMS, Peak, Variance, and Frequency Energy analysis on an ESP32 microcontroller provides a practical, scalable, and efficient solution for modern water management systems. The automatic pump shutdown feature adds an additional safety layer, making the system suitable for deployment in unattended environments.

The methodology successfully bridges the gap between computationally expensive ML models and simple threshold-based systems, achieving reliable detection with minimal hardware resources. Future enhancements including IoT connectivity and TinyML integration will further expand the system's capabilities and deployment scope.

16. REFERENCES

1.[1] Datta, S., Sarkar, S., et al., "Pipeline Leakage Detection Using Acoustic Emission and Machine Learning Algorithms," Sensors Journal, MDPI, Vol. 23, 2023.

2.[2] Abdel-Basset, M., et al., "Smart Buildings: Water Leakage Detection Using TinyML," Sensors Journal, MDPI, Vol. 23, 2023.

3.[3] Zhou, Y., Zhang, L., et al., "Pipeline Leak Detection Using Acoustic Emission and Sequential Deep Learning," Smart Cities Journal, 2024.

4.[4] Hunaidi, O., Wang, A., "Leakage Localisation Using Cross Correlation for Wireless Vibration Sensors in Pipeline Networks," NRC, Canada.

5.[5] Pham, V., Nguyen, T., "IoT Sensors to Monitor Pipeline Pressure and Flow Rate Combined with Machine Learning Algorithms," IEEE Access, 2023.

6.[6] Atobe, T., Iwase, M., "Embedded Systems for Real-Time Signal Processing," Springer, 2022.

7.[7] Oppenheim, A.V., Schafer, R.W., "Discrete-Time Signal Processing," 3rd ed., Prentice Hall, 2009.

8.[8] Espressif Systems, "ESP32 Technical Reference Manual," Version 5.0, 2023. [Online] Available: https://www.espressif.com

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